

Matrix Operation

$$\begin{array}{ccc} \boxed{a_{11}} & \boxed{a_{12}} & \boxed{a_{13}} \\ \boxed{a_{21}} & \boxed{a_{22}} & \boxed{a_{23}} \\ \boxed{a_{31}} & \boxed{a_{32}} & \boxed{a_{33}} \end{array} \times \begin{array}{ccc} \boxed{b_{11}} & \boxed{b_{12}} & \boxed{b_{13}} \\ \boxed{b_{21}} & \boxed{b_{22}} & \boxed{b_{23}} \\ \boxed{b_{31}} & \boxed{b_{32}} & \boxed{b_{33}} \end{array} = \begin{array}{ccc} \boxed{c_{11}} & \boxed{c_{12}} & \boxed{c_{13}} \\ \boxed{c_{21}} & \boxed{c_{22}} & \boxed{c_{23}} \\ \boxed{c_{31}} & \boxed{c_{32}} & \boxed{c_{33}} \end{array}$$

C_{22} MAC Arithmetic Operation

$$\boxed{a_{21}} \times \boxed{b_{12}} + \boxed{a_{22}} \times \boxed{b_{22}} + \boxed{a_{23}} \times \boxed{b_{32}} = \boxed{c_{22}}$$

Binary Accumulation & Fault

$$\begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 1 \\ \hline \end{array} + \begin{array}{|c|c|c|c|} \hline 0 & 0 & 1 & 0 \\ \hline \end{array} + \begin{array}{|c|c|c|c|} \hline 0 & 1 & 1 & 0 \\ \hline \end{array} = \begin{array}{|c|} \hline 9 \\ \hline \end{array}$$

Bit Flip

Correct: 7